

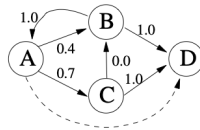
Roughly: read about existing, analyze, and propose a new outline / addition to a community trust system like advogatos user trust system for user verification.

Articles:

- **"Attack-Resistant Trust Metrics" (Raph Levien, 2003):** I believe the text outlining the original Advogato model. It characterizes the "group trust metric" and evaluates its resistance to spam and "trolls" in real-world community deployments.
- **"A group trust metric for identifying people of trust in online social networks" (ResearchGate, 2012):** This study proposes an "extended Advogato" metric that uses capacity-first maximum flow methods to improve the discovery of reliable users while preventing malicious access.
- **"Trust and Reputation Model for Various Online Communities" (StarTrust, 2011):** This paper introduces **StarTrust**, a model that addresses the "tyranny of the majority" and "echo chambers" by balancing global reputation with personalized local trust.
- **"Why Johnny Can't Encrypt: A Usability Evaluation of PGP 5.0" (Whitten & Tygar, 1999):** A seminal paper on the failure of the PGP web of trust. It argues that security fails when the user interface is too complex for average users to manage public-key encryption.
- **"Controversial Users demand Local Trust Metrics: an Experimental Study on Epinions.com" (Massa & Avesani, 2005):** This research highlights the failure of global reputation scores to handle users who are trusted by some but distrusted by others, proposing **MoleTrust** as a local alternative. (Great paper to take inspiration on and concur such a prevalent problem of trolling/disagreements on contr. characters)

Key Terms:

- Global Trust Metrics vs Local Trust Metrics - The formal definition of a global trust metric is hence $T : U \rightarrow [0,1]$ while local trust metrics are defined as $T : U \times U \rightarrow [0,1]$. Global is total community opinion while local is subjective user by user
- Controversial Users
- Trust Statements (Scores) are subjective, $T(A,B) = 0$ shows A completely distrusts B and $T(C,B) = 1$ shows C completely trusts A.



- Trust Network -
- Trust Metrics (inferring/predicting trust) - If A doesn't have a metric towards C but A trusts B and B trusts C there is inferred trust between A and C

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Step 1:
Input: source_user, trust_net, trust_prop_horizon
dist = 0; users[dist] = source_user
while (dist ≤ trust_prop_horizon) do
  dist ++
  users[dist] = users reachable from users[dist - 1]
  and not yet visited
  keep edges from users[dist - 1] to users[dist]

Step 2:
Output: trust_scores for users
dist = 0; trust(source_user) = 1
while (dist ≤ trust_prop_horizon) do
  dist ++
  foreach u in users[dist]
    
$$\text{trust}(u) = \frac{\sum_{i \in \text{pred}(u)} (\text{trust}(i) * \text{edge}(i, u))}{\sum_{i \in \text{pred}(u)} (\text{trust}(i))}$$


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Figure 2: MoleTrust pseudocode. $\text{pred}(u)$ returns predecessors p of user u for which $\text{trust}(p) \geq 0.6$. $\text{edge}(i, u)$ is the value of the statement issued by i on u .

- MoleTrust -
- controversy percentage, as $\# \text{trust} - \# \text{distrust} / \# \text{trust} + \# \text{distrust}$.
- EBay user trust (global) - $\text{trust}_{\text{ebay}}(\text{user}) = \# \text{trust} / \# \text{trust} + \# \text{distrust}$.
- PageRank
- Leave-One-Out Evaluation Technique

Ideas:

- Investigate Google Page Ranks solutions of looking at bounce time to verify users
- Analyze original advogado system (The Advogato Flow Model (Raph Levien))
- Analyze ebay // analyze review structure

Question 1:

Under this honors contract, I will conduct an in-depth literary and technical analysis of decentralized verification and trust systems that rely on collective community behavior rather than external identifiers such as government IDs, institutional roles, or centralized authorities. The project will examine existing models of trust, reputation, and verification, such as community-based vetting systems, reputation economies, and trust-weighted participation frameworks, analyzing their theoretical foundations, strengths, limitations, and failure modes. Building on this analysis, I will develop a basic conceptual framework for a community-derived trust verification system. This framework will outline how trust may be accumulated, challenged, reinforced, or degraded over time through user interactions, and under what conditions such a system is

appropriate or inappropriate to deploy. The project will also explore ethical implications, misuse scenarios, and structural weaknesses such as collusion, bias amplification, or exclusionary effects. The final deliverable will be a structured research paper that synthesizes existing scholarship with original analysis, proposes a coherent trust system design, and critically evaluates its feasibility, scope of use, and limitations. Where appropriate, the paper may reference illustrative models or simulations to clarify how the system would function in practice.

Question 2:

The main goal of this contract is to understand how trust can be built and checked in decentralized systems, especially in situations where there's no central authority or traditional identity system. This topic pulls from computer science, ethics, sociology, and system design. The outcome will be a clear, well-reasoned paper that shows I understand both the technical side and the bigger picture, along with a proposed trust framework that thoughtfully evaluates existing approaches and where they fall short.

Question 3:

This project deserves honors credit because it moves beyond the standard curriculum. Instead of just following instructions or using existing solutions, I am conducting independent research and developing my own original concepts. The "honors" element is defined by a shift from passive learning to critical analysis. I am not just studying how trust systems work; I am critically evaluating them: pinpointing exactly where these systems succeed, where they break down, and the specific reasons behind those failures.

Question 4:

Weeks 1–4: Research and reading on existing community-based trust and verification systems. Weeks 5–7: Compare systems, identify strengths and weaknesses, and outline the proposed approach. Weeks 8–10: Develop the trust framework and begin drafting the paper. Weeks 11–13: Write, revise, and tighten the paper for clarity and coherence. Weeks 14–15: Final edits and submission. Evaluation and grading will be based on: - Overall coherence of the paper - Clear evidence of engagement with and understanding of the material - Quality of analysis and conclusions